

Callable Bermudan Cross-Currency Swap

Quantitative model, simulation, exercise, and validation

Tim Glauner
TG Investments and Research, LLC

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Base model

Two Hull-White rate factors, lognormal FX, and Bermudan exercise by least-squares Monte Carlo (LSM).

- Domestic money-market account is the numeraire.
- Values and exercise comparisons are in domestic currency.
- Deterministic forecast/discount basis is frozen in version 1.

Core control

An independently calculated non-callable XCCY cash-flow PV is the primary benchmark and control variate.

Holder perspective

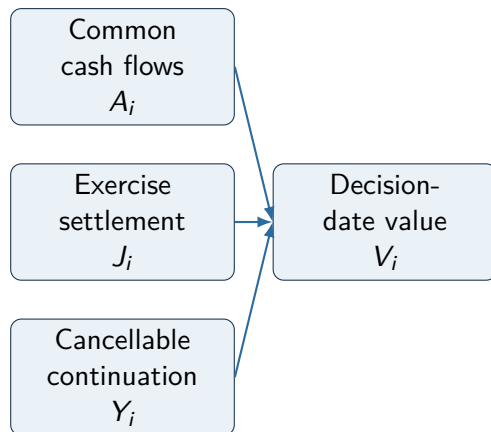
Positive cash flows are receipts by the exercise holder.

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- Domestic and foreign coupon legs.
- Optional initial and final notional exchanges.
- Exercise dates $t_1 < \dots < t_M$ with explicit settlement timing.
- FX quote S_t : units of DOM per one unit of FOR.

FX conversion at payment

$$CF_f(T) \mapsto S_T CF_f(T).$$



Continuation estimate

$$C_i = \mathbb{E}^{\mathbb{Q}^d}[Y_i \mid \mathcal{F}_{t_i}], \quad V_i = A_i + \max(J_i, C_i).$$

- Exercise when $J_i > C_i + \varepsilon_{\text{ex}}$.
- Exclude common A_i flows from both alternatives.
- Add A_i outside the comparison.
- Zero-fee cancellation: $J_i = 0$.
- Cancel when continuation is negative.
- Entry option: J_i is the value of entering the swap plus settlement.

Implementation consequence

The first exercise removes only the cancellable tail; paid, common, and delayed cash flows remain.

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$$\begin{aligned}dr_d(t) &= [\theta_d(t) - a_d r_d(t)] dt + \sigma_d(t) dW_d(t), \\dr_f(t) &= [\theta_f(t) - a_f r_f(t) - \rho_{fS} \sigma_f(t) \sigma_S(t)] dt + \sigma_f(t) dW_f(t), \\ \frac{dS_t}{S_t} &= [r_d(t) - r_f(t)] dt + \sigma_S(t) dW_S(t).\end{aligned}$$

Quanto adjustment

The negative foreign-rate drift adjustment follows from the change to the domestic pricing measure for the stated FX quotation.

No-arbitrage check

$$S_0 P_f^*(0, T) = \mathbb{E}^{\mathbb{Q}^d} [D_d(0, T) S_T].$$

Deterministic multi-curve mapping

For currency c , freeze

$$B_c(0, T) = \frac{P_c^F(0, T)}{P_c^D(0, T)},$$

$$P_c^F(t, T) = P_c^D(t, T) \frac{B_c(0, T)}{B_c(0, t)}.$$

- One stochastic rate factor per currency.
- Stochastic tenor or cross-currency basis is out of scope.

Instantaneous correlation

$$R = \begin{pmatrix} 1 & \rho_{df} & \rho_{dS} \\ \rho_{df} & 1 & \rho_{fS} \\ \rho_{dS} & \rho_{fS} & 1 \end{pmatrix}$$

$$R = R^T, \quad R \succeq 0.$$

Govern estimates, validate positive semidefiniteness, and stress all three correlations independently.

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- ① **Curves:** bootstrap domestic collateral and foreign effective curves; reprice helpers and enforce FX-forward parity.
- ② **Domestic HW:** calibrate to normal-vol swaptions on the exercise diagonal; constrain a_d and regularize $\sigma_d(t)$.
- ③ **Foreign HW:** apply analogous identification controls.
- ④ **FX diffusion:** fit $\sigma_S(t)$ to selected vanillas in the full hybrid model.
- ⑤ **Correlation:** combine governed historical/implied estimates with PSD checks and stresses.

Persist the evidence

Parameters, helper prices, relative errors, optimizer status, and curve/vol/correlation snapshot IDs.

Event grid contains every

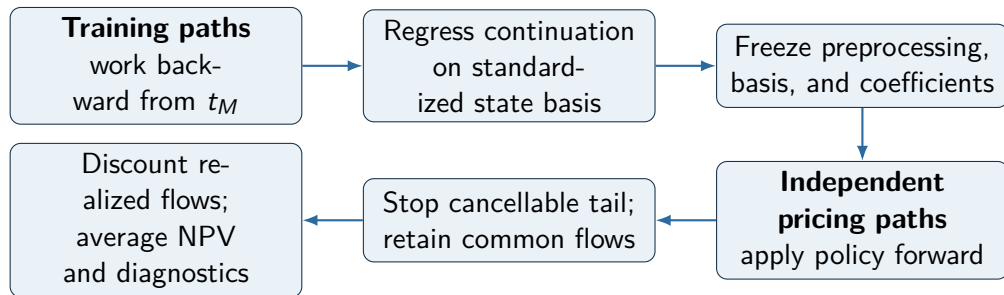
Fixing, accrual, payment, notional exchange, decision, effective, and settlement date, plus capped intermediate steps.

- Generate correlated normals from a factorization of R .
- Use exact conditional Gaussian Hull-White transitions when available.
- Integrate domestic short rates consistently for discounting.

Log-FX evolution

$$d \log S_t = \left[r_d(t) - r_f(t) - \frac{1}{2} \sigma_S^2(t) \right] dt + \sigma_S(t) dW_S(t).$$

- Apply analytic conditional bond values where possible.
- Start with antithetics; validate low-discrepancy variants separately.



Regression evidence

Record rank, condition number, coefficients, path count, and residual diagnostics. Re-train across seeds and compare policies on one common pricing sample.

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Report

Callable and non-callable NPV, embedded option value, Monte Carlo confidence interval, exercise probabilities, expected exercise time, per-leg PV, and calibration/regression diagnostics.

$$V_{\text{option}} = V_{\text{callable}} - V_{\text{noncallable}} \geq 0$$

Bump-and-revalue with common random numbers

- Domestic and foreign curve DV01.
- Forecast- and cross-currency-basis risk.
- Rate vegas and FX delta/gamma/vega.
- Three correlations and Hull-White parameter stresses.

State whether every bump recalibrates the model and retrains the policy.

- ① No exercise dates reproduce the non-callable XCCY PV.
- ② Zero-volatility and identical-currency limits reproduce deterministic and single-currency results.
- ③ One exercise date matches an independent European or nested-MC benchmark.
- ④ Domestic-discounted domestic assets and FX-converted foreign assets are martingales under $\mathbb{Q}^d$.
- ⑤ Prices converge across paths, time grids, regression bases, and seeds.
- ⑥ Multi-exercise cases bracket the LSM lower bound with a dual upper bound or a trusted nested-MC benchmark.
- ⑦ Differences use paired pathwise or covariance-aware errors under common random numbers.

Acceptance principle

Model fit alone is insufficient: exercise logic, event ordering, policy stability, numerical convergence, and limiting cases must all pass.

Version 1 delivers

A transparent three-factor hybrid reference model with controlled calibration, independent LSM policy evaluation, and auditable outputs.

Requires a new model version

Stochastic basis, smile dynamics beyond benchmark fitting, additional rate factors, collateral switching, or richer exercise rights.

Primary references

- Longstaff and Schwartz (2001), “Valuing American Options by Simulation.”
- QuantLib documentation: <https://www.quantlib.org/docs.shtml>.

Source note

This deck summarizes the TGIR callable Bermudan XCCY quantitative model article dated August 30, 2026.

Callable Bermudan XCCY Model

Questions and discussion

TG Investments and Research, LLC